Java Programming

WEEK 4 – Capstone Project

Focus: Build a complete Java application using OOP, collections, and file handling.

Topics Covered:

- MVC Pattern in Java
- Java Database Connectivity (JDBC)
- Mini GUI with JavaFX or Swing (Optional)
- Integrating modules (OOP + File + DB)
- Final testing and documentation

Capstone Project Options:

1. 🏫 **Student Management System (with Database)**
    - Modules: Add, Update, Delete, View Students
    - Backend: JDBC + MySQL Database
    - Store student records persistently
2. 💳 **ATM Simulation Project (Advanced)**
    - Features: PIN Login, Withdraw, Deposit, Check Balance, Transaction History
    - Store data in files or MySQL database
3. 🏠 **Hotel Booking Management System**
    - Add/Cancel bookings, calculate total cost, store data using ArrayList or DB

🎥 **Suggested Video:** Java Project Tutorial – CRUD Application